

Result:
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Neptun ID:	Draft papers: ☐
Name: _____	

The test consists of **4 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

- 1.) (essay)
- 2.) (essay)
- 3.)

a	b	c	d
☐	☐	☐	☐
- 4.)

☐	☐	☐	☐
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Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) What formula labels the root of the construction tree?

Score: 2 / 2.

2.) Define the conjunctive normal form (CNF).

Score: 2 / 2.

3.) Which statements are true for the following three formula?

$$A \Leftarrow \neg q, \quad B \Leftarrow q \vee \neg(q \supset p), \quad C \Leftarrow p \wedge (q \supset q)$$

(a) $A \Leftrightarrow B$,

-1 $\{A, B, C\}$ is satisfiable,

0 $A, B \models C$,

2 $(A \vee B)$ is valid.

4.) Which statements are true for the following three formula?

$$A \Leftarrow q \wedge q \vee q, \quad B \Leftarrow \neg(\neg q \wedge q), \quad C \Leftarrow p \vee p$$

(a) $A \Leftrightarrow B$,

2 $\{A, B, C\}$ is satisfiable,

(c) $A, B \models C$,

2 $(A \vee B)$ is valid.